Urban Mobility Optimization System (UMOS): System Design and Requirements Analysis

Natalie Mariño Figueroa - 20232020143
Dept. of Computer Engineering
Universidad Distrital Francisco José de Caldas
Email: nmarinof@udistrital.edu.co

Marlon Yecid Riveros Guio - 20231020208
Dept. of Computer Engineering
Universidad Distrital Francisco José de Caldas
Email: myriverosg@udistrital.edu.co

Edilson Santiago Sepulveda Cortes - 20231020237
Dept. of Computer Engineering
Universidad Distrital Francisco José de Caldas
Email: essepulvedac@udistrital.edu.co

Juan Esteban Cañon Solorza - 20232020078
Dept. of Computer Engineering
Universidad Distrital Francisco José de Caldas
Email: jecanons@udistrital.edu.co

***Abstract*—This document presents the design of the Urban Mobility Optimization System (UMOS), focused on improving transportation efficiency and user experience for the community of the Universidad Distrital Francisco José de Caldas. Based on the analysis conducted in Workshop #1, critical inefficiencies were identified related to congestion, long waiting times, low reliability, and mobility inequities. The proposed design integrates real-time information, intelligent route optimization, and adaptive mechanisms to respond to dynamic conditions such as demand fluctuations and environmental disruptions. The design is complemented with a modular system architecture, an implementation strategy based on microservices and cloud technologies, and explicit mechanisms for risk management, performance assurance, and scalability.**

***Index Terms*—Urban mobility, route optimization, multimodal systems, real-time information, requirements analysis, transportation equity**

I. Executive Summary

This document presents the system design of the Urban Mobility Optimization System (UMOS), aimed at improving transportation efficiency and user experience for the community of the Universidad Distrital Francisco José de Caldas. Based on the analysis conducted in Workshop #1, the current mobility system was identified as inefficient, highly variable, and sensitive to external conditions. Critical issues include congestion during peak hours, long commuting and waiting times, low reliability, and mobility inequities affecting users from different geographic areas.

The proposed system design addresses these challenges through the integration of real-time information, intelligent route optimization, and adaptive mechanisms capable of responding to dynamic conditions such as demand fluctuations and environmental disruptions. By leveraging multimodal transportation data and user-centered design principles, the system aims to improve decision-making processes for users, reduce uncertainty during travel, and enhance overall system performance. The expected outcome is a more reliable, efficient, scalable, and equitable mobility system that supports sustainable urban transportation.

II. Requirements Analysis

A. Synthesis of Workshop #1

The Urban Mobility Optimization System (UMOS) was analyzed as a complex and dynamic system composed of multiple interconnected subsystems, including physical transport infrastructure, fleet and operations, digital information systems, and governance mechanisms.

The analysis revealed that the system exhibits non-linear behavior and strong dependency on demand patterns, particularly during peak hours. Demand concentration within short time windows leads to congestion, increased waiting times, and system overload; in critical periods, transport capacity is exceeded, resulting in delays and reduced service quality.

Environmental factors, especially rainfall, significantly impact system performance by increasing waiting times, reducing the availability of alternative transport modes, and amplifying congestion. Additionally, the system demonstrates emergent behaviors such as demand amplification, feedback loops, and adaptive user decision-making. Several structural bottlenecks were identified, including mixed-traffic corridors, limited transfer nodes, and centralized access points, which create vulnerabilities and reduce system resilience. Overall, the system was evaluated as underperforming in terms of efficiency, reliability, and equity, highlighting the need for a more robust and adaptive design approach.

B. System Objectives

The primary objective of the proposed system design is to improve urban mobility efficiency by optimizing route planning and reducing overall travel time for users. Additional objectives include:

- Improving system reliability under dynamic and uncertain conditions.
- Reducing congestion during peak demand periods.
- Enhancing access to accurate and real-time mobility information.
- Supporting user decision-making through intelligent recommendations.

- Reducing variability in travel times and system performance.
- Promoting equitable access to transportation services.
- Enabling scalability and adaptability for future system growth.

C. System Constraints

The system operates under multiple constraints derived from the analysis, which must be considered in the design process:

- **Infrastructure constraints:** Limited road capacity and mixed-traffic corridors reduce transport efficiency and increase delays.
- **Demand constraints:** High demand concentration during peak hours creates system overload conditions.
- **Structural constraints:** Dependence on limited transfer nodes introduces single points of failure.
- **Environmental constraints:** Weather conditions, particularly rainfall, significantly affect system performance.
- **Regulatory constraints:** Policies such as fixed fare systems and traffic restrictions influence system dynamics.
- **Technological constraints:** Limitations in real-time data availability and system integration affect information accuracy.

These constraints define the operational limits of the system and guide the design of feasible and realistic solutions.

D. Stakeholder Needs

UMOS must address the needs and expectations of multiple stakeholder groups identified in Workshop #1 and the requirements report.

- **Students:** Require fast, affordable, and reliable transportation options, with clear and real-time information about routes, travel times, and disruptions.
- **Faculty and administrative staff:** Need predictable, safe, and consistent commuting conditions that support fixed work schedules.
- **Transport operators:** Seek efficient demand management, optimized routing, and improved operational performance through better data and planning tools.
- **Institutional stakeholders:** Expect solutions aligned with university mobility policies, sustainability goals, and long-term campus planning.
- **Urban mobility authorities:** Require data-driven insights on flows, bottlenecks, and user behavior to support traffic management and policy decisions.

Balancing these needs is essential to ensure system effectiveness, adoption, and long-term sustainability.

III. System Architecture

A. Overview

The architecture of UMOS follows a modular, microservices-based design that reflects the subsystem decomposition obtained in the system analysis (physical transport, fleet operations, information systems, and governance). The software system is primarily situated in the information subsystem but interacts closely with the others through real-time data feeds, operational constraints, and regulatory policies. [file:1][file:6][file:7]

At a high level, the architecture is organized into seven logical modules: trip capture, external data integration, route optimization engine, notification management, user-facing presentation layer, persistence and analytics, and policy and rules management. Each module is implemented as one or more microservices that communicate using REST/gRPC APIs and event streams, ensuring loose coupling and fault isolation.

B. Context Diagram

The context diagram (to be included as Figure 1) depicts UMOS as a central system interacting with human actors (students, faculty, staff) and external systems (TransMilenio and SITP open data, SDM traffic feeds, weather services, university academic system). Incoming flows include trip requests and real-time data; outgoing flows include optimized routes, alerts, and aggregated reports for institutional and governmental stakeholders. This diagram clarifies the system boundaries and enumerates all external dependencies that must be addressed by interfaces and risk mitigation strategies.

C. Diagrama de Contexto

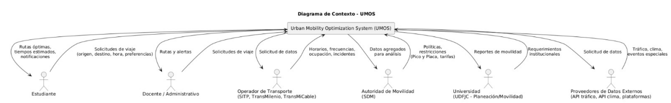

Fig. 1. UMOS system context diagram

D. Logical Architecture Diagram

The logical architecture diagram (Figure 2) decomposes UMOS into its main modules and illustrates their responsibilities and interactions. The trip capture module receives origin, destination, time, and user preferences and forwards standardized trip requests to the route optimization engine. The data integration module consolidates transit schedules, vehicle positions, traffic conditions, weather information, roadworks, and academic schedules. The route engine consumes these inputs together with policy constraints to compute optimal and alternative multimodal routes, and stores results in the persistence layer. The notification module subscribes to relevant events and pushes alerts to clients, while the presentation layer exposes the information through web and mobile interfaces.

E. Arquitectura Lógica

F. Component Diagram

The component diagram (Figure 3) refines each logical module into concrete software components such as *TripRequestController*, *RoutingEngine*, *TrafficDataAdapter*, *WeatherDataAdapter*, *NotificationService*, *MobilityDB*, and *PolicyEngine*. Each component exposes well-defined interfaces (for example, *computeOptimalRoute()*, *getTrafficSnapshot()*, *sendAlert()*) and depends only on the contracts of other components, which maps closely to NestJS modules in the

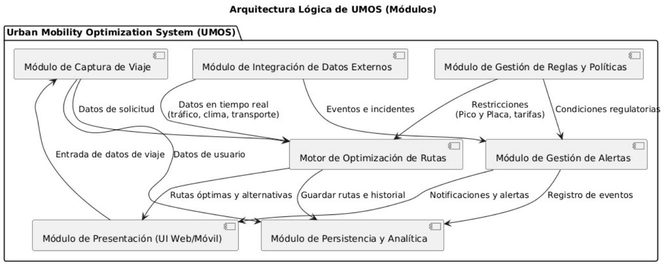

Fig. 2. UMOS logical architecture diagram

implementation. This view supports maintainability and testability by making explicit which component is responsible for fulfilling each functional requirement.

G. Diagrama de Componentes

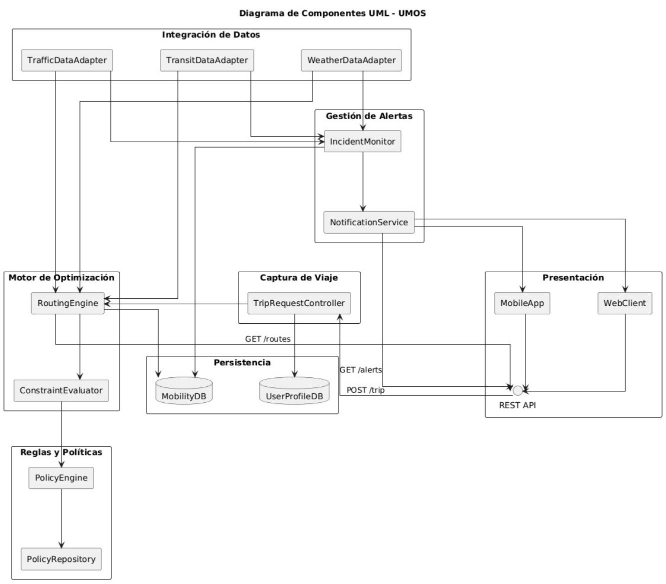

Fig. 3. UMOS component diagram

H. Sequence and Data Flow Diagrams

For the primary use case "Plan Trip", a sequence diagram (Figure 4) shows how a trip request flows from the client application to the backend services and back. The user submits a request through the mobile or web interface; the *TripRequestController* validates and records it; the *RoutingEngine* queries the data integration adapters for transit, traffic, and weather data; applies policies via the *PolicyEngine*; stores route options in *MobilityDB*; and responds with optimal and alternative routes, including travel time estimates. The user then subscribes to updates via the notification service, which pushes alerts if incidents or disruptions occur along the chosen path. This dynamic view demonstrates that all functional requirements related to routing, real-time updates, and notifications are covered by the architecture.

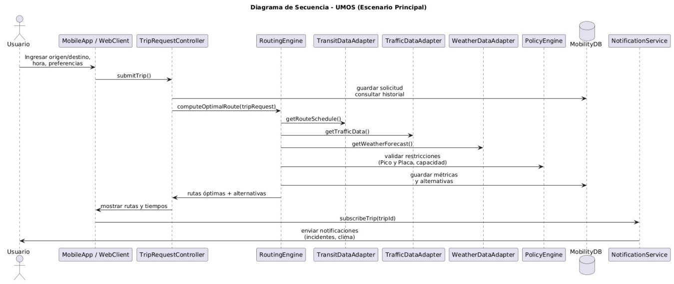

Fig. 4. Sequence diagram of the "Plan Trip" use case

I. Diagramas de Secuencia

J. Deployment Diagram

The deployment diagram (Figure 5) assigns components to physical or virtual nodes in the target infrastructure. Client applications run on smartphones and browsers and communicate with an API gateway deployed in Google Kubernetes Engine (GKE). Backend microservices execute in a Kubernetes cluster, while PostgreSQL/PostGIS, Redis, and TimescaleDB run as managed database services. External systems (TransMilenio, SDM, weather services, academic system) are represented as separate nodes connected via secure API calls. This layout enables horizontal scaling of critical services, blue-green deployments, centralized monitoring, and enforcement of security and privacy controls.

K. Diagrama de Despliegue

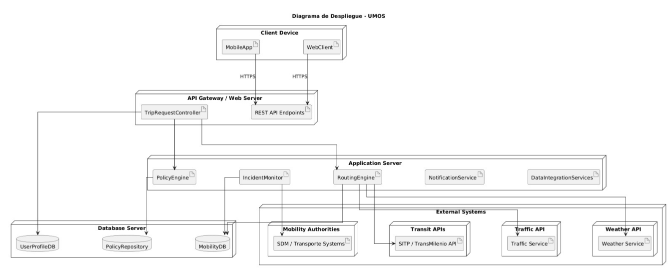

Fig. 5. UMOS deployment diagram

IV. Technical Implementation Strategy

The implementation strategy builds on the Workshop 2 design report and targets an Agile–Scrum process with DevOps practices. The system is decomposed into microservices implemented in Node.js/TypeScript with NestJS, exposing REST and WebSocket interfaces, and orchestrated on GKE. The mobile client is built with React Native and Expo, using Mapbox GL for multimodal route visualization. Data storage combines PostgreSQL/PostGIS for geospatial data, Redis for caching, and TimescaleDB for time-series analytics. [file:6]

Development is organized into four phases over 20 weeks: (1) foundation (infrastructure, CI/CD, user management), (2) data integration (GTFS, traffic, weather, academic schedule), (3) route engine and mobile MVP, and (4) advanced features and hardening (prediction models, personalization, load testing, security). Each phase is aligned with semester constraints and the priority matrix derived from Workshop 1, ensuring early delivery of high-impact capabilities such as rain-aware routing and real-time data integration.

V. Risk and Complexity Management

External API Dependency

UMOS consumes real-time data from five critical external sources—TransMilenio (GTFS-RT), Secretaría Distrital de Movilidad (SDM), weather services, roadworks updates, and the academic system—none of which guarantee stable versioning or consistent uptime. To mitigate this, the architecture implements a Kong 3.x API Gateway with the circuit breaker pattern. If an external API fails or exceeds latency thresholds, the system automatically serves cached data from Redis 7 for up to 60 seconds, preventing the route optimization engine from blocking or returning incomplete results.

Single Point of Failure Propagation

The physical bottleneck at Universidades station can disrupt mobility for nearly a third of the university community during peak hours. To avoid replicating this fragility in software, UMOS adopts a microservices architecture that isolates functional modules so that failures in non-critical services—such as notification or analytics—do not impact the core routing engine. This separation ensures that essential trip planning functionality remains available even under partial degradation.

Environmental Disruptions

Rainfall has been shown to increase waiting times by 63 % and significantly reduce the availability of alternative modes, directly impacting predictive accuracy and GPS reliability. UMOS addresses this by integrating weather data as a first-class input into the routing engine, allowing dynamic adjustment of recommendations based on current and forecasted conditions and triggering mode-shift suggestions when cycling becomes impractical. [file:1][file:6][file:7]

Data Freshness and User Trust

Survey results indicate that 39 % of users are dissatisfied with the timeliness of information. To address this, UMOS implements a data transparency mechanism that displays the age of each information element (for example, “Updated 45s ago”), managing expectations and providing implicit feedback on data source reliability and system performance. [file:1][file:7]

Fallback Modes

In the event of complete loss of GTFS-RT feeds, UMOS degrades gracefully to static GTFS schedules provided by TransMilenio, preserving basic routing capabilities at reduced accuracy rather than becoming unavailable. During periods of high demand or partial data loss, the system prioritizes core routing over ancillary features such as personalization or analytics reporting, aligning with the principle of graceful degradation.

VI. Performance and Quality Assurance

The performance and quality strategy for UMOS is designed to address the current system’s low reliability score by defining quantitative targets and implementing automated mechanisms for scalability, availability, and continuous validation. [file:6][file:7]

Scalability Under Peak Demand

Demand concentration during the critical morning peak (07:30–08:15) results in approximately 128 arrivals every 15 minutes at Sabio Caldas campus. To handle this load, UMOS employs Kubernetes Horizontal Pod Autoscaler (HPA) to dynamically replicate the route optimization engine during peak periods and scale down after 09:30 to respect cost constraints (about $780 USD per month covered by academic credits). The system is engineered to process around 14,000 I/O events per minute, with the routing engine scaling independently from user management and notification modules.

Availability and Deployment Strategy

UMOS targets 99.5 % uptime through blue-green deployments that maintain two identical production environments. New versions are deployed to the inactive environment, validated with automated smoke tests, and then activated via traffic switching at the API gateway; failures trigger instantaneous rollback with no user impact. Deployments are scheduled in off-peak windows identified in the variability analysis to further reduce risk. [file:1][file:6][file:7]

Code Quality, Testing, and Security

Quality assurance is embedded into the development lifecycle through ESLint with Airbnb rules, strict TypeScript typing, peer review of pull requests, and explicit code coverage targets (80 % for critical modules, 60 % for supporting services). Continuous integration pipelines execute unit, integration, and performance regression tests on every change. Security and privacy are enforced via OAuth 2.0 with short-lived JWTs, TLS 1.3 for data in transit, AES-256 encryption for data at rest, and a 48-hour retention policy for raw location traces, in compliance with Colombian data protection law.

VII. Derived Requirements

Based on the findings from Workshop #1 and previous system analysis, the following requirements are defined. [file:7]

A. Functional Requirements

- The system must allow users to input origin and destination locations.
- The system must generate optimal routes based on real-time traffic conditions.
- The system must integrate multiple transportation modes (bus, BRT, bicycle, walking).
- The system must provide real-time updates on traffic, delays, and disruptions.
- The system must suggest alternative routes when congestion is detected.

- The system must estimate travel time for each route option.
- The system must notify users about incidents, road closures, or environmental disruptions.
- The system must support multimodal trip planning and route combinations.

B. Non-Functional Requirements

- **Usability:** The system should provide a simple, intuitive, and user-friendly interface.
- **Performance:** The system should deliver route recommendations within a few seconds.
- **Reliability:** The system must provide accurate and consistently updated information.
- **Availability:** The system should be accessible at all times (24/7).
- **Accessibility:** The system should be compatible with mobile devices and different user conditions.
- **Security:** The system must ensure the protection of user data and privacy.

C. Performance, Scalability, Reliability, and Equity Requirements

- The system should reduce average waiting times and travel durations compared to the current baseline.
- The system must maintain acceptable performance under peak demand conditions.
- The system must support a large and growing number of users and new transportation modes and data sources.
- The system must operate under varying conditions such as congestion, demand fluctuations, and weather changes, minimizing variability in travel time predictions and maintaining stable operation despite partial failures.
- The system should reduce mobility inequities among users from different geographic areas and provide equal access to information and optimization services.

D. Design Foundation

These requirements are directly derived from the system analysis conducted in Workshop #1 and are complemented by subsequent modeling and design activities. Together they establish the foundation for the system architecture and guide implementation strategies aimed at improving performance, resilience, and user experience.

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